

Fire — Inductive Encoding

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The system: elemental combustion — exothermic oxidation with self-sustaining flame front propagation through a fuel medium.

D — What is the dimensionality of the state space?

Combustion fields (temperature, species concentration, velocity) are continuum field-theoretic. The degrees of freedom are not finite discrete states but infinite-dimensional field configurations. $\rightarrow D_\infty$

$$\langle D_\infty; ?; ?; ?; ?; ?; ?; ?; ?; ?; ?; ? \rangle$$

T — What is the topology of the interaction?

The flame front is a genuine crossing point: fuel and oxidizer streams do not merely branch — they intersect and exchange across the front. The double-well ignition landscape (unlit / burning) and the crossing of reactant and product flows make $T_\bowtie$ the correct topology, not a simple branching network. $\rightarrow T_\bowtie$

$$\langle D_\infty; T_\bowtie; ?; ?; ?; ?; ?; ?; ?; ?; ?; ? \rangle$$

R — What is the relational mode?

Heat release rate and reaction progress form an adjoint pair: one drives the other functorially. This is $R_\dagger$ (adjoint / one-directional functorial mapping from field to rate), not the symmetric bidirectional feedback of R_{lr} . $\rightarrow R_\dagger$

$$\langle D_\infty; T_\bowtie; R_\dagger; ?; ?; ?; ?; ?; ?; ?; ?; ? \rangle$$

P — What is the parity / symmetry structure?

At the molecular level, radical chain propagation involves quantum superposition of reaction pathways and transition states. The symmetry is not classical $\mathbb{Z}_2$ but the quantum superposition symmetry $P_\psi \rightarrow P_\psi$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; ?; ?; ?; ?; ?; ?; ?; ? \rangle$$

F — What is the fidelity?

Macroscopic flame phenomenology is classical. The critical ignition threshold does not require quantum coherence to describe. $\rightarrow F_\ell$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; F_\ell; ?; ?; ?; ?; ?; ?; ? \rangle$$

K — What is the kinetic character?

Flame propagation lives on the slow manifold of near-equilibrium critical dynamics. The flame front advances on the timescale set by the slowest reactive mode — not fast relaxation, not frozen. $\rightarrow K_{\text{slow}}$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; ?; ?; ?; ?; ?; ? \rangle$$

G — What is the scope / granularity?

Radiative heat transfer and convective coupling in combustion are long-range and universal: the flame exchanges energy with fuel across all scales, not just nearest neighbors. $\rightarrow G_\mathbb{N}$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\mathbb{N}; ?; ?; ?; ?; ? \rangle$$

\Gamma — What is the interaction grammar?

The flame broadcasts thermal radiation to all surrounding fuel simultaneously in a one-to-many pattern. This is not conjunctive (all conditions required at once) but broadcast (one source, many receivers). $\rightarrow \Gamma_{\text{broad}}$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\mathbb{N}; \Gamma_{\text{broad}}; ?; ?; ?; ? \rangle$$

\Phi — What is the criticality?

Combustion is a self-sustaining critical phenomenon with a sharp ignition threshold — the paradigm case of Φ_c . $\rightarrow \Phi_c$

$$\langle D_\infty; T_\infty; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\mathbb{N}; \Gamma_{\text{broad}}; \Phi_c; ?; ?; ? \rangle$$

H — What is the temporal depth / chirality?

The critical ignition condition at the flame front is memoryless. Whether a given parcel of fuel ignites depends only on the instantaneous local temperature and concentration — not on the history of how that state was reached. $\rightarrow H_0$

$$\langle D_\infty; T_\boxtimes; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\aleph; \Gamma_{\text{broad}}; \Phi_c; H_0; ?; ? \rangle$$

S — What is the stoichiometry?

Many fuel and oxidizer species couple heterogeneously — the reaction network is many-to-many with unmatched coupling strengths. $\rightarrow n:m$

$$\langle D_\infty; T_\boxtimes; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\aleph; \Gamma_{\text{broad}}; \Phi_c; H_0; n:m; ? \rangle$$

Ω — What is the winding / topological protection?

Combustion has exactly two stable phases: burning and extinguished. These are not continuously deformable into each other — there is a $\mathbb{Z}_2$ topological invariant distinguishing them. The naive first-pass encoding (Ω_0 , trivial) missed this: fire is not merely a critical network, it is a topologically protected binary. $\rightarrow \Omega_{\mathbb{Z}_2}$

$$\langle D_\infty; T_\boxtimes; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\aleph; \Gamma_{\text{broad}}; \Phi_c; H_0; n:m; \Omega_{\mathbb{Z}_2} \rangle$$

0.1 Final Type

$$\langle D_\infty; T_\boxtimes; R_\dagger; P_\psi; F_\ell; K_{\text{slow}}; G_\aleph; \Gamma_{\text{broad}}; \Phi_c; H_0; n:m; \Omega_{\mathbb{Z}_2} \rangle$$

Tier: $O_2^\dagger$ — critical + $\mathbb{Z}_2$ -protected + unbounded domain (R5). **Crystal address:** 4428422 (cell 102, inner 22022). **C-score:** ≈ 0.59 (both gates open; K_{slow} on slow manifold, $T_\boxtimes$ at crossing topology).